CCC-C-447
OCTOBER 14, 1960
SUPERSEDING
Fed. Spec. CCC-P-191a
September 26, 1944

FEDERAL SPECIFICATION

CLOTH, COTTON, PERCALE

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE

1.1 This specification covers the requirements for one type of cloth, cotton, percale.

2. APPLICABLE SPECIFICATIONS, STANDARDS, AND OTHER PUBLICATIONS

2.1 The following specifications and standards, of the issues in effect on date of invitation for bids, form a part of this specification:

Federal Specifications:

CCC-T-191—Textile Test Methods.

PPP-P-51 — Packaging, Packing and Marking of Textile Fabrics (Woolens, Worsteds, Cottons, Silks and Synthetics).

Federal Standards:

Fed. Std. No. 4—Glossary of Fabric Imperfections.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications, Standards, and Handbooks and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, Seattle, and Washington, D. C.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications, Standards, and Handbooks from established distribution points in their agencies.)

Military Standards:

MIL-STD-105 — Sampling Procedures and Tables for Inspection by Attributes.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 **Other publications.** — The following document forms a part of this specification. Unless otherwise indicated the issues in effect on date of invitation for bids shall apply:

Textile Fiber Products Identification Act.

(Copies may be obtained free of charge from the Federal Trade Commission, Washington 25, D. C.)

3. REQUIREMENTS

3.1 **Standard sample.**—When a standard sample has been established, the cloth shall match that standard for shade and be equal to or better than the standard for colorfastness.

3.2 **Yarn.**—The cotton used in the manu-

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facture of the yarn shall be cleaned, carded, drawn and spun into singles yarn.

3.3 Color. — The color shall be bleached white or dyed as specified (see 6.2). The use of sulfur dyes or substances containing elementary sulfur compounds capable of oxidation to sulfuric acid is prohibited (see 4.4).

3.3.1 *Color matching.* — The color shall match the approved standard shade sample under natural (north sky) daylight or artificial daylight having a color temperature of 7500° Kelvin and shall be a good approximation to the standard shade under incandescent lamplight at 2800° Kelvin (see 6.3).

3.3.2 *Colorfastness.*

3.3.2.1 *Rating per standard sample.*—The dyed cloth shall show fastness to wet dry cleaning, crocking, laundering and light equal to or better than the standard sample. When no standard shade samples are available, the dyed cloth shall show good fastness to wet dry cleaning, crocking, laundering and light. Testing shall be as specified in 4.4.

3.4 Physical requirements.—The finished cloth shall conform to the requirements shown in table I when tested as specified in 4.4.

3.4.1 *Width.*—The width shall be as specified and shall be the minimum width exclusive of selvages (see 6.2).

3.5 Finish. — The cloth shall be singed, bleached or dyed and may be calendered. The residual size, finishing or nonfibrous material content shall not exceed 6 percent when tested as specified in 4.4.

3.6 Shrinkage.—The cloth shall be preshrunk and shall not shrink more than 1 percent in either the direction of the warp or the filling when tested as specified in 4.4. The preshrinking process used shall not be identified on the cloth, the ticket, or on the package.

3.7 Length. — Unless otherwise specified, the cloth shall be in one continuous piece of not less than 40 yards (see 6.2) and shall be put-up in flat folded pieces.

3.8 Labeling.—Each roll or flat fold piece shall be labeled, ticketed or invoiced in accordance with the Rules and Regulations of the Textile Fiber Products Identification Act dated March 3, 1960.

3.9 Workmanship. — The cloth shall conform to the quality and grade of product established by this specification and the occurrence of defects shall not exceed the applicable acceptable quality levels.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Contractors inspection responsibility. —Unless otherwise specified herein, the supplier is responsible for the performance of all inspection requirements prior to submission for Government inspection and accept-

TABLE I.—*Physical requirements*

Weight per square yard (minimum)	Texture (yarns per inch minimum)		Breaking strength grab method minimum		Weave
	Warp	Filling	Warp	Filling	
Ounces			*Pounds*	*Pounds*	
3.0	85	72	45	30	Plain

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Glossary number Fed. Std. No. 4	Defects	Classification	
		Major	Minor
90, 91, 92, 55	Hole, cut or tear, including selvage	X	
78	Smash, any	X	
33	Broken or missing end:		
	Three or more contiguous regardless of length	X	
	—two contiguous, missing more than 3 inches		X
	—single, missing more than 9 inches		X
4, 10, 11	*Broken picks, missing picks, mispicks, double picks:*		
	—resulting in two or more contiguous double picks	X	
	or		
	—a distinct absence of three or more contiguous picks	X	
	or		
	—light filling bar or light, tender or weak place	X	
	or		
	—harness skips repeating over part or all of the width	X	
	or		
	—resulting in two or more double picks in any linear yard, extending 4 inches or more		X
9, 12, 35	Open place or crack, opening at widest point three or more yarn diameters, or sequence of openings resulting in a weak spot	X	
	Clearly visible*, but opening at widest point less than three yarn diameters		X
35	*Hitchback:*		
	—distorted area 2 inches or more in combined warp and filling directions	X	
	—distorted area less than 2 inches in combined warp and filling directions		X
13, 34	*Floats or skips:*		
	—multiple 1 inch or more in either warp or filling directions	X	
	—less than 1 inch in either warp or filling directions		X
	—single over 2 inches or more or sequence of pin floats		X
1, 2	Heavy filling bar or heavy place clearly visible*		X
3, 5	*Jerked-in filling, slough-off, slub, slug or gout:*		
	—more than 1/16 inch in thickness and more than 1 inch in length	X	
	—more than 1/16 inch in thickness and 1 inch or less in length		X
	—more than 1/32 inch but not more than 1/16 inch in thickness and more than 4 inches in length		X
5, 27, 77	Bunchy filling—An accumulation of four or more slubs (smaller in size than indicated above clearly visible*) occuring in the same 2 linear inches		X
69, 70, 75, 76, 74	*Spot, stain or streak:*		
	—clearly visible* 1/2 inch or more in combined length and width dimensions	X	
	—clearly visible* more than 1/4 inch but less than 1/2 inch in combined length and width dimensions		X

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Glossary number Fed. Std. No. 4	Defects	Classification	
		Major	Minor
6, 28	Kinks or knots (oversized and extending from the surface of the cloth) four or more in 1 linear yard of cloth		X
40, 41, 36	Wrong draw, reed mark—mixed end clearly visible*		X
42	Temple mark—resulting in holes or frayed fibers, clearly visible*		X
89	Tender or weak place	X	
59, 61	Wrinkle or crease—embedded		X
51	Scalloped selvage, deepest indentation more than ½ inch		X
	Tight selvage—clearly noticeable slackness or pucker in body of cloth along selvage		X

* Clearly visible at normal inspection distance (approximately 3 feet).

ance. Except as otherwise specified, the supplier may utilize his own facilities or any other commercial laboratory acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order.

4.2 Examination of the end item.—Except when otherwise specified by Civil Government agencies, inspection shall be in accordance with the provisions of Military Standard MIL–STD–105 except where otherwise indicated. Defects found during the examination shall be classified in accordance with list shown in 4.2.1, 4.2.2, and 4.2.3.

4.2.1 *Yard-by-yard examination.*—The required yardage of each piece shall be inspected on the face and visual defects classified as listed. The defects found shall be counted regardless of their proximity to each other except where two or more defects represent a single local condition of the cloth, in which case only the more serious defects shall be counted. A continuous defect shall be counted as one defect for each warpwise yard or fraction thereof in which it occurs. The unit of product shall be 1 linear yard. The acceptable quality level (AQL) shall be 2.5 major or 10.0 total defects per hundred units. The sample size shall be based upon inspection level II of Military Standard MIL–STD–105. The lot size shall be expressed in units of 1 yard each. The number of pieces from which the sample is to be selected shall be in accordance with table II herein. An approximately equal number of yards shall be examined in each piece of the sample.

4.2.2 *Overall examination.* — Each defect listed shall be counted not more than once in each piece examined. The unit of product shall be one piece. The number of pieces selected for yard-by-yard examination shall comprise the sample for this examination and the number of defects acceptable shall be as specified in table II.

Defects

Overall uncleanness
Uneven weaving throughout piece
Overall neppiness
Width less than minimum specified
Selvage curled, folded, rolled, slack or loopy, continuously or intermittently over more than half the piece
Colors—off shade, shaded, or non-uniform, poorly penetrated or mottled

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TABLE II.—*Sample size*

Lot size in yards	Sample size in pieces	Acceptance number
Up to 1,300[1]	3	0
1,301 to 3,200	5	0
3,201 to 8,000	7	0
8,001 to 22,000	10	0
22,001 to 110,000	15	1
110,001 to 550,000	25	1
550,001 and up	50	2

[1] If a lot contains fewer than three pieces, each piece in the lot shall be examined.

4.2.3 *Examination for length.*

4.2.3.1 *Individual pieces.*—The piece shall be examined for gross length. Any gross length found to be less than 40 continuous yards or more than 2 yards below the gross length marked on the ticket shall be considered a defect with respect to length. The unit of product for this examination shall be one piece. The number of pieces selected as sample (sample size) for this examination and the acceptance number shall be in accordance with table II.

4.2.3.2 *Total yardage in sample.*—The lot shall be unacceptable if the total of the actual gross lengths of the pieces in the sample is less than the total of the gross lengths marked on the tickets.

4.3 Examination of preparation for delivery requirements.—An examination shall be made to determine that packaging, packing and marking requirements of Section 5 of this specification are complied with. Defects shall be scored in accordance with the list below. The sample unit shall be one shipping container fully packed, selected just prior to the closing operation. Defects of closure listed below shall be examined on shipping containers fully prepared for delivery. The lot size shall be the number of containers in the inspection lot. The inspection level shall be L–4 and the AQL shall be 4.0 defects per hundred units.

Examine	Defect
Marking (exterior and interior)	Omitted, incorrect, illegible, of improper size, location, sequence, or method of application.
Materials	Any component missing. Any component damaged, affecting serviceability.
Workmanship	Inadequate application of components, such as: incomplete closure of case liners, container flaps, loose strapping, inadequate stapling. Bulging or distortion of containers.
Weight (exterior and interior)	Gross, net weight exceeds requirements.

4.4 Testing of the end item.—The methods of testing specified in Federal Specification CCC–T–191, wherever applicable, and as listed in table III herein shall be followed. The physical and chemical values specified in Section 3 apply to the average of the determinations made on a unit of product for test purposes as specified in the applicable test method. The unit of product shall be 2 yards, full width, of the finished cloth. Except when otherwise specified by Civil Government Agencies the sample size and acceptance number shall be as follows:

Lot size (yards)	Sample size	Acceptance number
800 or less	2	0
801 to 22,000	3	0
22,001 and over	5	0

4.4.1 *Test for presence of sulfur.*—Dissolve 100 g. of stannous chloride crystals in 100 ml. of hydrochloric acid (35 percent concentration) and add 50 ml. of distilled water. To preclude interference in dye identification, sample first shall be boiled in distilled water to remove any trace of chemical used in the dyeing process. One to 1.5 grams of the finished material, cut into small pieces, shall be used for the test. Place the sample in a test tube of suitable size and cover with the stan-

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nous chloride solution. Place a filter paper which has been moistened with one drop of lead acetate solution (made by dissolving 5 grams of reagent grade lead acetate to 100 cc. of distilled water; add a drop of glacial acetic acid to clear up if cloudy), over the mouth of the test tube. Heat slowly over a low flame until the solution is boiling. Solution shall not be boiled for more than 10 to 15 seconds as prolonged boiling will reduce the sensitivity of the test. A brown to black stain appearing on the paper indicates the presence of sulfur.

5. PREPARATION FOR DELIVERY

(Federal Standard No. 102 shall be referred to for definitions and applications of the various levels of packaging and packing for civil agency procurements).

5.1 Packaging.—Packaging shall be level A, B, or C as specified (see 6.2 and 6.2.1).

5.1.1 *Levels A, B, and C.*—The cloth shall be flat folded and packaged in accordance with the applicable requirements for class 3 fabrics of Federal Specification PPP–P–51.

5.2 Packing.—Packing shall be level A, B, or C as specified (see 6.2 and 6.2.1).

TABLE III.—*Test methods*

Characteristic	Requirement reference	Test method	Number of determinations	Reported as
Weight	Table I	5041	5	Average of the 5 determinations to the nearest 0.1 ounce
Fiber identification	3.2	1200	1	Pass or fail
Weave	Table I	Visual	1	Pass or fail
Threads per inch (warp and filling)	Table I	5050	5	Average of the 5 determinations to the nearest whole number
Breaking strength (warp and filling)	Table I	5100	5	Average of the 5 determinations to the nearest pound
Test for presence of sulfur	3.3	4.4.1	1	Satisfactory or unsatisfactory
Colorfastness to:				
Laundering	3.3.2	5610	1	When rated per no standard sample: excellent, good, fair or poor. When rated per standard sample: satisfactory or unsatisfactory
Light[1]		5660	1	
Wet dry cleaning		5622	1	
Crocking		5651	1	
Nonfibrous material[2]	3.5	2611	2	Average of the 2 determinations to the nearest 0.1 percent
Shrinkage (warp and filling)	3.6	5550	3	Average of the 3 determinations not beyond the 1st decimal point

[1] In the event of a dispute resulting from the test with method 5660, or as a result of suspected anomalous behavior of certain dye types or formulations, the contracting officer may authorize the exposure to natural light in accordance with method 5662.

[2] The nonfibrous content shall be determined in accordance with method specified and shall be calculated from the following formula:

$$\text{Nonfibrous content, percent} = \frac{O - S}{O} \times 100$$

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5.2.1 *Levels A, B, and C.*—The cloth shall be packed in accordance with the applicable requirements of Federal Specification PPP–P–51.

5.3 **Marking.**—In addition to any special marking required by the contract or order, shipments shall be marked in accordance with the applicable requirements of Federal Specification PPP–P–51.

6. NOTES

6.1 **Intended use.**—The cloth, cotton, percale covered by this specification is intended for use in the manufacture of flags and brassards.

6.2 **Ordering data.** — Procurement documents should specify the following:

(*a*) Title, number, and date of this specification.
(*b*) Color of cloth required (see 3.3).
(*c*) Width of cloth required (see 3.4.1).
(*d*) Length of piece if other than specified in 3.7.
(*e*) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.2.1 In the preparation of contracts or orders, it must be noted that various options, choices, and alternatives as indicated in Federal Specification PPP–P–51 may be exercised in the preparation of delivery of cloth.

6.3 **Standard sample.**—For access to standard sample, address the procuring office issuing the invitation for bids.

6.4 **Transportation description.** — The transportation description and minimum weights applicable to this commodity are:

Rail:
Cotton piece goods not otherwise indexed by name in the original piece.
Carload minimum weight — Any quantity.

Motor:
Cotton piece goods not otherwise indexed in the original piece.
Motor volume minimum weight — Any quantity.

Notice. — When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded, by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

MILITARY INTEREST:
Army—M O Q
Navy—S MC
Air Force—MAAMA.

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